

Dr. B. C. Roy Engineering College, Durgapur

Department of Electrical Engineering
B. Tech (EE) | 5th Semester EE2 | Autonomy

Subject: Power Electronics Subject Code: EE-504

Flip Learning – Pre-Class Self-Study Note

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i Topics Covered in This Note:

- ✔ **Topic 1:** Ideal vs. Practical Switch
- ✔ **Topic 2:** Turn-On and Turn-Off Times
- ✔ **Topic 3:** Switching Losses
- ✔ **Topic 4:** Two-Transistor Model of SCR
- ✔ **Topic 5:** Regenerative Turn-On Mechanism of SCR

CO: CO1 Module: 1 (Introduction) and Module 2 (PNPN Devices) Section: EE-2, B2024-28

References: M. H. Rashid, *Power Electronics*, PHI, 3rd Ed., Ch. 1 and Ch. 4; P. S. Bimbhra, *Power Electronics*, Khanna Publishers, 3rd Ed., Ch. 2–3.

NPTEL: Dr. G. Narayanan, IISc Bangalore – <https://nptel.ac.in/courses/108/108/108108035/>

A Important – Read This Before the Flip Learning Session:

This note covers topics from Module 1 and Module 2 not yet lectured in full. The in-class MCQ on **1st September 2026, 2:10–3:00 PM** is based **entirely** on this material. No theory will be taught in class.

⌚ Estimated reading time: 40–50 minutes 📅 Shared: 30-August-2026 (Sunday, evening)

1 ⚡ Topic 1 – Ideal vs. Practical Switching Characteristics

1.1 The Ideal Switch – What We Wish For

An ideal power switch has three perfect properties:

- **ON state:** Zero voltage drop ($V_{on} = 0$), so zero conduction loss.
- **OFF state:** Zero leakage current ($I_{off} = 0$), so zero blocking loss.
- **Switching:** Instantaneous transition ($t_{on} = t_{off} = 0$), so zero switching loss.

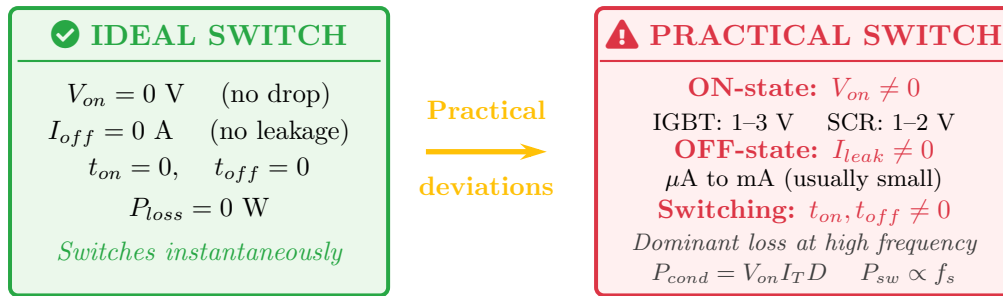


Figure 1: Ideal vs. practical power switch – the three key deviations.

1.2 Why Each Deviation Matters

Deviation	Loss Formula	Significance	GATE Tip
ON-state drop	$P_{cond} = V_{on} I_T D$	Proportional to duty ratio and current	Dominant at low f_s
Leakage current	$P_{leak} = V_{off} I_{leak}$	Usually negligible (%)	Often ignored in calculations
Finite t_{on}, t_{off}	$P_{sw} = \frac{1}{2} V_{off} I_{on} (t_{on} + t_{off}) f_s$	Grows linearly with f_s	Dominant at high f_s

2 🕒 Topic 2 – Turn-On Time and Turn-Off Time

2.1 Turn-On Time: $t_{on} = t_d + t_r$

$t_{on} = t_d + t_r$

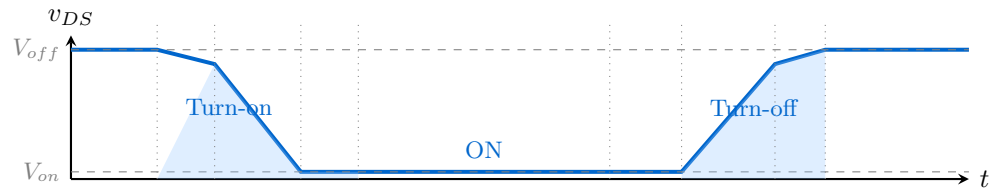
t_d = **delay time** (gate capacitance charging; current at 0%–10%) t_r = **rise time** (current rises from 10% to 90% of rated value)

2.2 Turn-Off Time: $t_{off} = t_s + t_f$

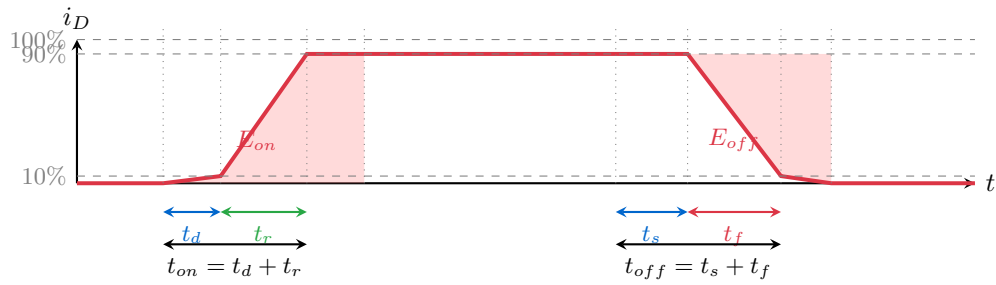
$t_{off} = t_s + t_f$

t_s = **storage time** (minority carriers clearing; current stays at 100%) t_f = **fall time** (current drops from 90% to 10%)

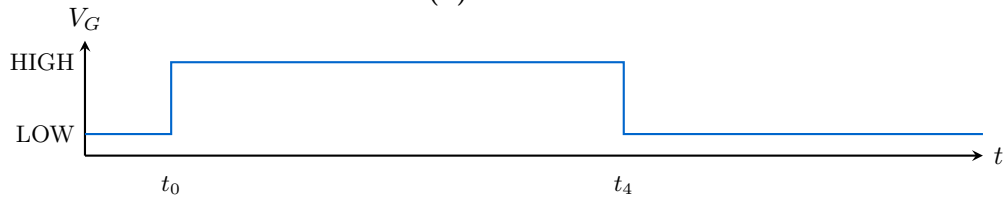
Switching loss = $\int v(t) \cdot i(t) dt$ (shaded area)



(c) Device voltage



(b) Device current



(a) Gate signal

Figure 2: Complete switching timing diagram for a power device.

💡 **Key Distinction – Storage Time t_s :** t_s is unique to **bipolar minority-carrier devices** (BJT, IGBT, SCR/Thyristor). During t_s , the gate signal is already removed but the device still conducts because excess minority carriers stored in the base/drift region have not yet cleared. Power MOSFETs (majority-carrier devices) have no t_s – they are therefore much faster to turn off. This is why MOSFETs dominate high-frequency (> 100 kHz) converters.

3 Topic 3 – Switching Losses

3.1 The Switching Loss Formula

$$E_{on} = \frac{1}{2} V_{off} I_{on} t_{on} \quad E_{off} = \frac{1}{2} V_{off} I_{on} t_{off}$$

$$P_{sw} = (E_{on} + E_{off}) f_s = \frac{1}{2} V_{off} I_{on} (t_{on} + t_{off}) f_s$$

Total device loss: $P_{total} \approx P_{cond} + P_{sw}$ (leakage is negligible)

⚠ Critical Insight – Linear in f_s : $P_{sw} \propto f_s$. Doubling the switching frequency doubles the switching loss. This is the fundamental trade-off in converter design: higher frequency allows smaller magnetic components, but increases device losses.

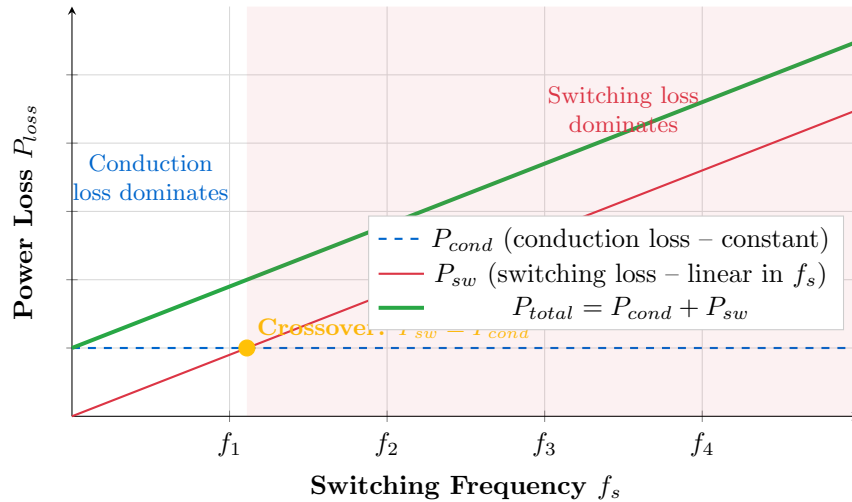


Figure 3: Device loss vs. switching frequency.

🔑 Maximum Switching Frequency Constraint:

$$f_{s,max} \lesssim \frac{1}{t_{on} + t_{off}}$$

If the switching period T_s is not much larger than $(t_{on} + t_{off})$, the device spends a significant fraction of each cycle in transition – efficiency collapses. In practice, $T_s \geq 5-10 \times (t_{on} + t_{off})$ is used as a design rule.

4 📖 Topic 4 – Two-Transistor Model of the SCR

4.1 SCR Structure – The PNPN Device

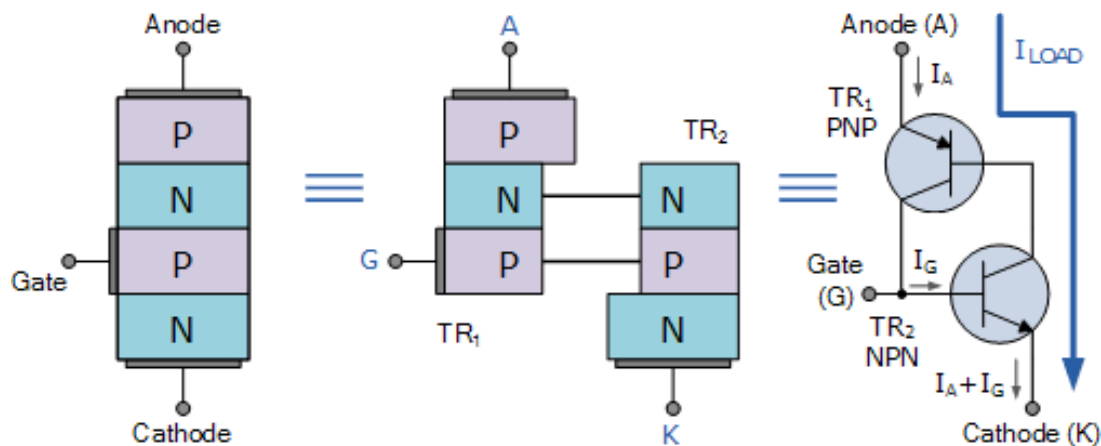


Figure 4: PNPN layer structure and two-transistor equivalent (from your image).

4.2 BJT Circuit-Level Two-Transistor Model

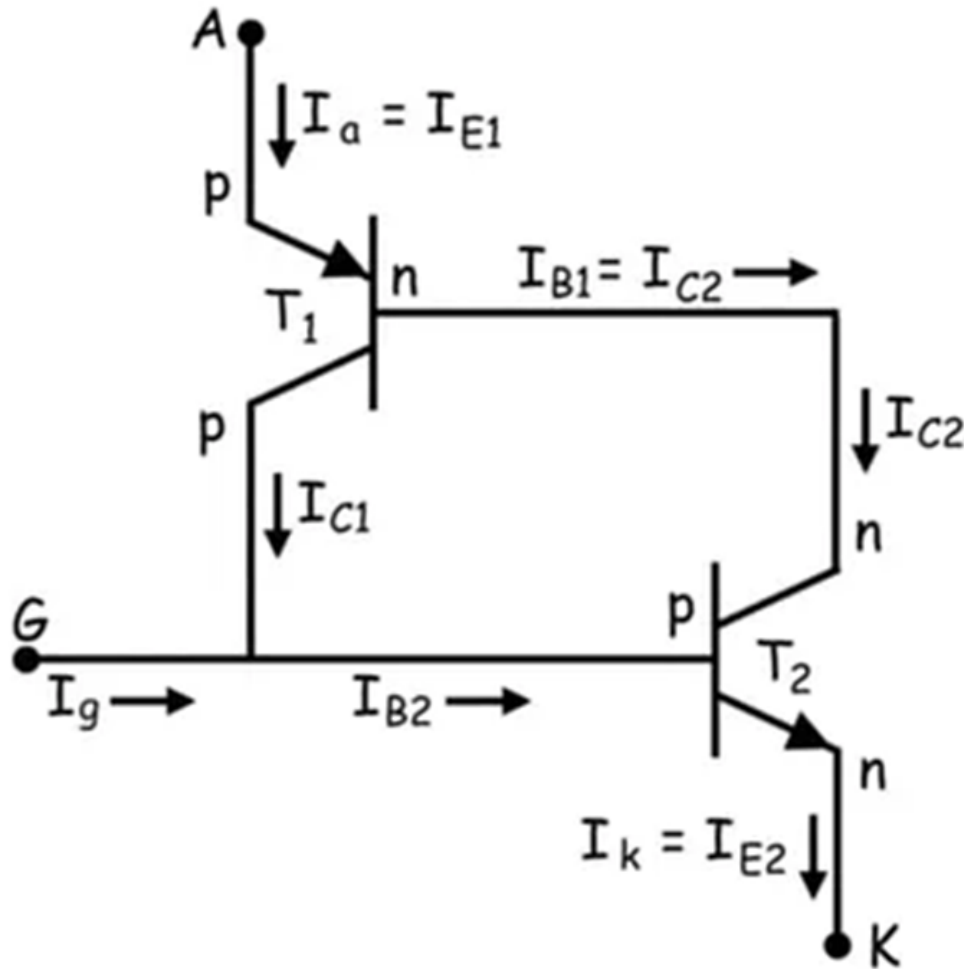


Figure 5: Circuit-level realisation of the two-transistor model (from your image).

4.3 Full Mathematical Derivation of I_A

≡ Derivation from BJT circuit equations – step by step

Step 1 – Define currents using common-base BJT model:

$$I_{C1} = \alpha_1 I_{E1} + I_{CBO1}, \quad I_{C2} = \alpha_2 I_{E2} + I_{CBO2}$$

Step 2 – Express emitter currents:

$$I_{E1} = I_A, \quad I_{E2} = I_K = I_A + I_G$$

Step 3 – Apply cross-coupling constraint:

$$I_A = I_{C1} + I_{C2} + I_{CBO1} + I_{CBO2}$$

Step 4 – Substitute BJT expressions:

$$I_A = \alpha_1 I_A + \alpha_2 I_A + \alpha_2 I_G + I_{CBO1} + I_{CBO2}$$

Step 5 – Solve for I_A :

$$I_A = \frac{I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

Step 6 – Physical interpretation:

Condition	Denominator	Effect
$\alpha_1 + \alpha_2 \ll 1$	$1 - (\alpha_1 + \alpha_2) \approx 1$	Small I_A – blocking
$\alpha_1 + \alpha_2 \rightarrow 1$	Denominator $\rightarrow 0$	$I_A \rightarrow \infty$ – latch-on
$\alpha_1 + \alpha_2 = 1$	$= 0$	Latch condition, gate no longer needed
$I_G = 0$	Depends on V_{AK}	Breakover when avalanche raises sum to 1

5 Topic 5 – Regenerative Turn-On Mechanism

5.1 Step-by-Step Turn-On Process

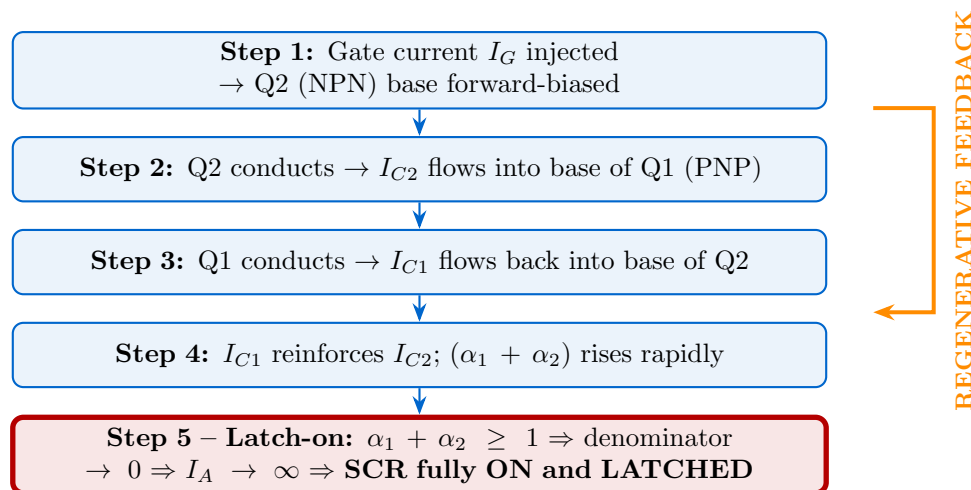


Figure 6: Step-by-step regenerative turn-on of an SCR.

✔ Gate Pulse – Short Duration

Once $\alpha_1 + \alpha_2 \geq 1$, gate can be removed. A pulse of just **5–20 μs** is enough. Internal feedback sustains ON state.

🔒 Latch and Turn-Off

Latch: $\alpha_1 + \alpha_2 \geq 1$

Stay ON: $I_A > I_H$ (holding current)

Turn OFF: Reduce $I_A < I_H$ via commutation. Gate **cannot** turn off the SCR.

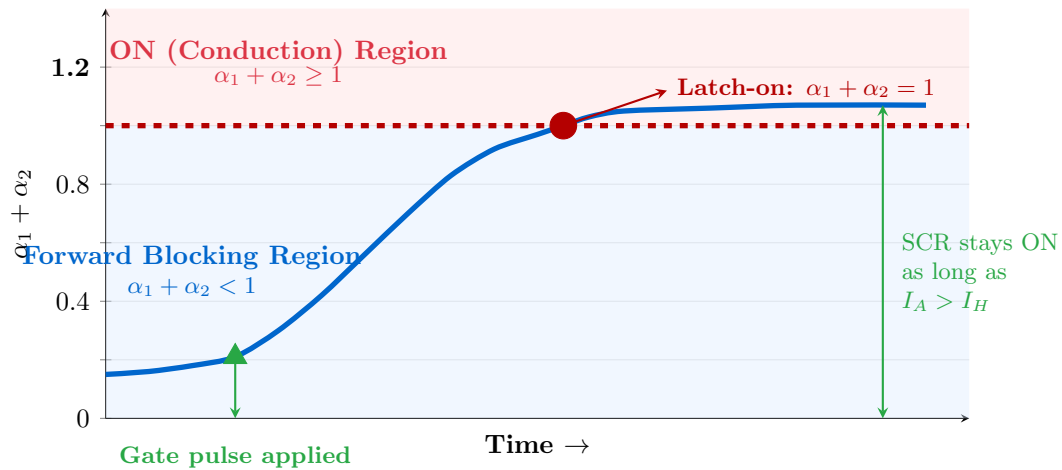


Figure 7: $\alpha_1 + \alpha_2$ vs. time during SCR turn-on.

5.2 Forward Breakover – The Uncontrolled Firing Risk

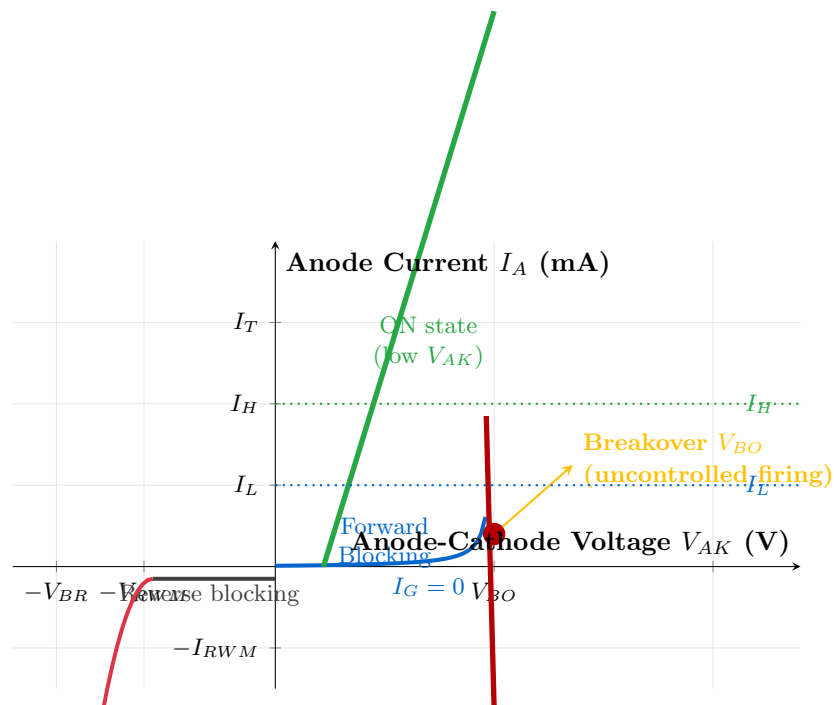


Figure 8: SCR V-I characteristic.

5.3 SCR Turn-On Methods – Summary

Turn-On Method	Physical Mechanism	Controllable?
Gate triggering	Inject I_G into Q2 base; initiates regenerative feedback	YES (normal)
Forward breakover	V_{AK} exceeds V_{BO} ; avalanche raises $\alpha_1 + \alpha_2$ to 1 without gate	NO (fault)
dv/dt triggering	Rapid V_{AK} rise; displacement current $C_{J2} dv/dt$ acts as gate current	NO (undesirable)
Temperature	High T raises leakage and α values	NO (fault)
Light (LASCR)	Photon-generated carriers trigger turn-on in the gate region	YES (LASCR only)

Table 1: SCR turn-on methods.

6 Quick Revision Summary – All 5 Topics Complete Revision Table – Study this before the MCQ

Topic	Key Formula / Concept	GATE/ESE Focus Point
Ideal switch	$V_{on} = 0, I_{off} = 0, t_{on} = t_{off} = 0, P_{loss} = 0$	Does not exist – benchmark only
Turn-on time	$t_{on} = t_d + t_r; t_r = 10\% \text{ to } 90\% \text{ of rated current}$	Standard data-sheet definition
Turn-off time	$t_{off} = t_s + t_f; t_s = \text{minority carrier storage}$	t_s only in bipolar devices (BJT, IGBT, SCR)
Switching loss	$P_{sw} = \frac{1}{2} V_{off} I_{on} (t_{on} + t_{off}) f_s$	Linear in f_s – doubles if f_s doubles
Max frequency	$f_{s,max} \lesssim 1/(t_{on} + t_{off})$	MOSFET fastest; SCR slowest
PNPN structure	4 layers P1-N1-P2-N2; junctions J1, J2, J3	J2 is the controlling junction
Two-transistor model	Q1 (PNP) + Q2 (NPN); $I_{C1} \rightarrow \text{base of Q2}; I_{C2} \rightarrow \text{base of Q1}$	Cross-coupled = positive feedback
I_A expression	$I_A = (I_G + I_{CBO1} + I_{CBO2})/(1 - \alpha_1 - \alpha_2)$	Denominator $\rightarrow 0$ when $\alpha_1 + \alpha_2 = 1$
Latch-on condition	$\alpha_1 + \alpha_2 \geq 1$	Gate pulse not required after latch
Turn-off of SCR	Reduce $I_A < I_H$ (holding current)	Gate has no control over turn-off
Breakover V_{BO}	$\alpha_1 + \alpha_2 = 1$ reached by avalanche at $V_{AK} = V_{BO}$	Undesirable; rating violation
Latching current I_L	Minimum I_A needed to sustain latch after gate removed	$I_L > I_H$ always

Table 2: Complete revision summary for Flip Learning MCQ.

⚠ Most Commonly Confused Pairs in GATE/ESE:

1. **Latching current I_L vs Holding current I_H :** I_L is needed to *sustain* latch right after gate removed. I_H is the minimum I_A for continued ON state. Always $I_L > I_H$.
2. **t_s vs t_f :** During t_s , current has *not yet started falling* (carriers still stored). During t_f , current *actually falls* from 90% to 10%.
3. **Gate triggering vs Breakover:** Gate triggering is controlled (desired). Breakover is uncontrolled (fault).
4. **Latch-on condition vs Turn-off:** $\alpha_1 + \alpha_2 \geq 1$ causes latch-on. To turn OFF, reduce $I_A < I_H$ – the gate cannot turn off an SCR.

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